

Table 59. DAC accuracy

Specified by design - not tested in production unless otherwise stated.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
DNL	Differential non linearity ⁽¹⁾	DAC output buffer ON	-2	-	2	LSB
		DAC output buffer OFF	-2	-	2	
	Monotonicity	10 bits	-	-	-	-
INL	Integral non linearity ⁽²⁾	DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ Ω	-4	-	4	LSB
		DAC output buffer OFF, $C_L \leq 50$ pF, no R_L	-4	-	4	
Offset	Offset error at code 0x800 ⁽²⁾	DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ Ω $V_{REF+} = 3.6$ V	-	-	± 12	LSB
		DAC output buffer OFF, $C_L \leq 50$ pF, no R_L	-	-	± 8	
Offset1	Offset error at code 0x001 ⁽³⁾	DAC output buffer OFF, $C_L \leq 50$ pF, no R_L	-	-	± 5	LSB
OffsetCal	Offset error at code 0x800 after factory calibration	DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ Ω $V_{REF+} = 3.6$ V	-	-	± 5	LSB
Gain	Gain error ⁽⁴⁾	DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ Ω	-	-	± 1	%
		DAC output buffer OFF, $C_L \leq 50$ pF, no R_L	-	-	± 1	
TUE	Total unadjusted error	DAC output buffer ON $C_L \leq 50$ pF, $R_L \geq 5$ k Ω	-	-	± 30	LSB
		DAC output buffer OFF $C_L \leq 50$ pF, no R_L	-	-	± 12	
TUECal	Total unadjusted error after calibration	DAC output buffer ON $C_L \leq 50$ pF, $R_L \geq 5$ k Ω	-	-	± 23	LSB
SNR	Signal-to-noise ratio ⁽⁵⁾	DAC output buffer ON $C_L \leq 50$ pF, $R_L \geq 5$ k Ω 1 kHz, BW 500KHz	-	67.8	-	dB
		DAC output buffer OFF $C_L \leq 50$ pF, no R_L 1 kHz, BW 500 KHz	-	67.8	-	
THD	Total harmonic distortion ⁽⁵⁾	DAC output buffer ON $C_L \leq 50$ pF, $R_L \geq 5$ k Ω , 1 kHz	-	-78,6	-	dB
		DAC output buffer OFF $C_L \leq 50$ pF, no R_L , 1 kHz	-	-78,6	-	
SINAD	Signal-to-noise and distortion ratio ⁽⁵⁾	DAC output buffer ON $C_L \leq 50$ pF, $R_L \geq 5$ k Ω , 1 kHz	-	67.5	-	dB
		DAC output buffer OFF $C_L \leq 50$ pF, no R_L , 1 kHz	-	67.5	-	
ENOB	Effective number of bits	DAC output buffer ON $C_L \leq 50$ pF, $R_L \geq 5$ k Ω , 1 kHz	-	10.9	-	bits
		DAC output buffer OFF $C_L \leq 50$ pF, no R_L , 1 kHz	-	10.9	-	

1. Difference between two consecutive codes minus 1 LSB.
2. Difference between the value measured at Code i and the value measured at Code i on a line drawn between Code 0 and last Code 4095.
3. Difference between the value measured at Code (0x001) and the ideal value.
4. Difference between the ideal slope of the transfer function and the measured slope computed from code 0x000 and 0xFFFF when the buffer is OFF, and from code giving 0.2 V and ($V_{REF+} - 0.2$ V) when the buffer is ON.
5. The signal is -0.5 dBFS with $F_{sampling} = 1$ MHz.

5.3.20 Comparator characteristics

Table 60. COMP characteristics

The input capacitance is negligible compared to the I/O capacitance.

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Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V _{DDA}	Analog supply voltage	-	2.70	-	3.60	V
V _{IN}	Comparator input voltage range	-	0	-	V _{DDA}	
V _{BG}	Scaler input voltage	-	(1)			
V _{SC}	Scaler offset voltage	-	-	±5	±10	mV